6 October 2026

The Editors
Advanced Science

Agentic design of continuous flow processes with chemical knowledge and laboratory constraints

Amirreza Mottafegh, Mincheol Park, Jun-young Jo, Jin Suk Myung, Boyoung Y. Park, and Gwang-Noh Ahn

Dear Editors,

We submit the above manuscript for consideration as a Research Article in Advanced Science. The study investigates how large language models and agentic reasoning can advance flow chemistry by translating a batch procedure into a connected experimental design. This task requires chemical knowledge to be reconciled with feed preparation, reactor configuration and the capabilities of the available equipment. We develop a multi-agent workflow that starts from a batch protocol, laboratory inventory and researcher objectives, then combines literature precedents, handbook-derived engineering guidance, quantitative calculations and specialist review to propose a complete flow process. Implemented as FlowPilot, the workflow connects feed compositions and stage conditions to specific equipment and a process diagram through a graphical interface. The broader advance is a practical way to organize language-model knowledge around the coupled decisions needed to turn established chemistry into a testable flow experiment.

Our evaluation spans a matched five-model benchmark, module-ablation and retrieval analyses, computational dry runs, and laboratory implementation of two connected reaction sequences. Across 90 benchmark designs, the mean assessed score increased from 0.794 for one-shot generation to 0.917 for the agentic workflow, while judge-reported critical flags decreased from 98 to 4. For each experimental response set, computational review of 12 candidates produced one selected design. The photoredox addition–oxidation sequence achieved 97–98% NMR yield and 94% isolated yield, compared with reported literature flow yields of 90% and 88%, respectively, and a batch isolated yield of 95%, under different conditions. A second, telescoped amidation afforded up to 86% NMR yield and 84% isolated yield. These results show that agent-selected starting configurations can be experimentally productive when assessed against literature procedures developed through experimental optimization. By linking language-model evaluation to apparatus design and chemical outcomes, the study should interest Advanced Science readers working across artificial intelligence, chemistry and engineering.

Sincerely,

Boyoung Y. Park and Gwang-Noh Ahn
Corresponding authors

Kyung Hee University | boyoungy.park@khu.ac.kr
Korea Research Institute of Chemical Technology | gnahn@krict.re.kr